

1. Image Enhancement in the Spatial Domain

Image Enhancement in the Spatial domain is the process of improving the quality of an image by directly modifying its pixel values. It makes images clear, improves contrast, reduces noise and highlights important details.

1. Basic Gray-level Transformations

- Modify the intensity of individual pixels
- Techniques include image negative, Contrast Stretching, gamma correction and thresholding.
- Use: Enhances brightness and contrast in dark or low quality images.

2. Histogram Processing

- A Histogram shows the distribution of pixel intensities
- Histogram Equalisation improves image contrast
- Use: Suitable for low-contrast images such as Medical.

3. Spatial Filtering

- Uses a small mask (kernel) to process neighbouring pixels
- Smoothing filters (Mean, Gaussian, Median) remove noise
- Sharpening filters (Laplacian, Sobel) enhance edges
- Use: Suitable for noisy or blurred images.

2. Image Smoothing vs. Image Sharpening

Difference b/w Image Smoothing and Image Sharpening

Image Smoothing	Image Sharpening
Removes noise & Small details	Enhances edges & Fine details
Produces a blurred image	Produces a clearer & Sharper image
Use low-pass filters	Uses high-pass filters
Ex: Mean, Gaussian	Ex: Sobel, Laplacian

Working Principle:-

1. Spatial Smoothing filters

- Operate using a small mask (kernel) over the image.
- Replace each pixel with the average or median of pixels.

2. Spatial Sharpening filters

- ~~Highlight~~ intensity changes b/w neighbouring pixels.
- Improve image clarity and detail.

Applications:-

- Image Smoothing: Used for removing noise in medical image, CCTV images and scanned documents.
- Image Sharpening: Used for enhance edges in Satellite images, fingerprints, text documents and medical image.

Image Enhancement in Frequency domain

Image enhancement in frequency domain improves quality by transforming the image from the spatial domain to the frequency domain using the Fourier Transform.

1. Low-Pass filters (LPF)

- Allow low-frequency components to pass and remove high-frequency components.
- Smooth the image and reduce noise.

Applications:

- Noise Removal
- Medical Imaging

2. High-Pass Filters (HPF)

- Enhance edges, fine details & object boundaries
- Increase image sharpness

Applications:

- Edge detection
- Satellite imaging

Comparison:

Low-Pass Filter	High-Pass Filter
• Removes noise • Produces smooth/ blurred images • Removes high-frequency components	• Enhances edges • Produces sharp images • Removes low-frequency components

4) Image Segmentation

Image Segmentation is the process of dividing an image into meaningful regions or objects for easier analysis and recognition.

Comparison of Segmentation Techniques

Technique	Working Principle	Advantages	Limitations	Application
Point detection	Detects isolated pixels with different intensity	Simple and Fast	Detects only individual points	Defect detection, Star detection
Line Detection	Detects horizontal, vertical or diagonal lines using mask	Identifies linear features	Sensitive to noise	Road detection, document
Edge Detection	Detects boundary where pixel intensity changes	Accurate object boundary detection	Affected by noise	Medical Imaging.
Thresholding	Separates object and background using intensity	Simple and computationally efficient	Not suitable for uneven lighting	Document scanning
Region-based Segmentation	Groups neighbour pixels with similar properties	Produces complete image regions	More computationally expensive	Medical image analysis

Most Appropriate Technique

- Edge detection
- Thresholding
- Region - Based Segmentation

Edge Detection, Edge linking and Boundary Detection

Introduction:

Edge Detection, edge linking and boundary detection are fundamental techniques in image processing.

1. Edge Detection:

It identifies points in an image where there is a sudden change in intensity.

Working Principle

- Computes intensity gradients using operators
- High gradient values indicate edges.

Example:

Detecting the edges of vehicles in traffic images.

2. Edge Linking

~~It~~ Connects broken or disconnected edge pixels to form continuous edges.

Working Principle:

- Uses pixel connectivity, gradient direction.
- Methods includes Hough Transform.

Example:

Connecting broken extracts the complete outline of an object using links.

Boundary Detection

Boundary detection extracts the complete outline of an object using linked edges.

Working principle

- Identifies the outer contour of object
- Uses Contour tracing or region-based Segmentation

Techniques.

Example:

Detecting the boundary of a tumour in MRI Scans

Contribution to Object Recognition & Defect detection

- Object recognition helps identify object shapes, size and position accurately.
- Defect detection detects cracks, scratches, dents and manufacturing defects in industrial products.
- Medical imaging identifies organs, tumours.
- Autonomous Vehicles detects road lanes, obstacles.